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3.3.6 Applying Q

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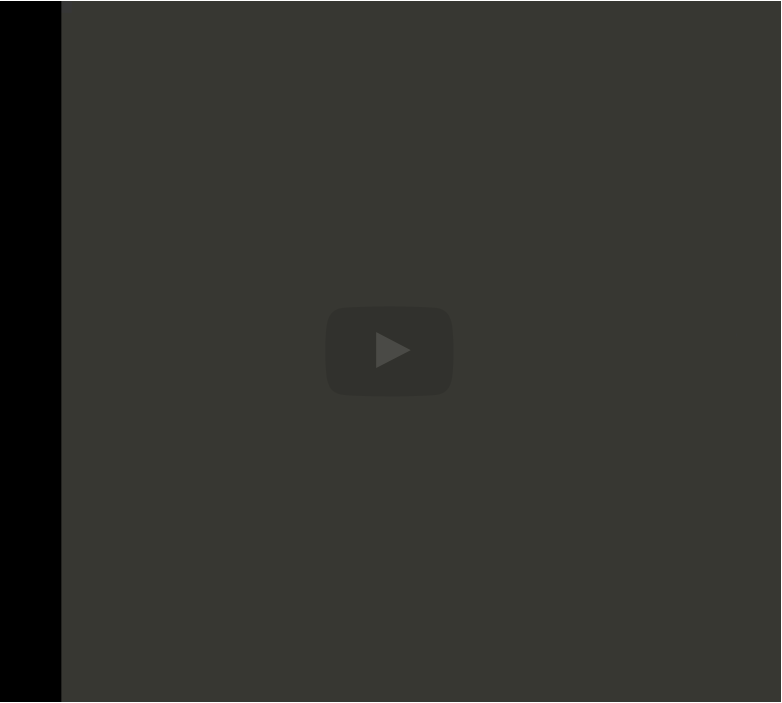
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matrix. And

the only part we really care about actually is this vector u_{21} which stores the

Householder transform that needs to be applied next.

And then similarly, you have a vector of all of these scalars, τ , that were

involved in all of these Householder transformations. So you need to pick out

the τ corresponding to the current Householder transformation. And then

you expose the different

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Reading assignment

1/1 point (graded)



Read Unit 3.3.6 of the notes. [\[LINK\]](#)

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? Minor typo/inconsistency in Fig 3.3.6.1.	2

Homework 3.3.6.1.

1/1 point (graded)

What is the approximate cost of algorithm in

```
[y] = Apply_QH(A, t, y)
A → ( ATL | ATR
      ABL | ABR ), t → ( tT
                        tB ), y → ( yT
                                yB )
ATL is 0 × 0 and tT, yT have 0 elements
while n(ABR) > 0
    ( ATL | ATR
      ABL | ABR ) → ( A00 | a01 A02
                      a10T | α11 a12T
                      A20 | a21 A22 ),
    ( tT
      tB ) → ( t0
              τ1 ), ( yT
                    yB ) → ( y0
                            ψ1
                            y2 )
    Update ( ψ1
            y2 ) := ( I - 1/τ1 ( 1
                                u21 ) ( 1 u21H ) ) ( ψ1
                                                    y2 )
    via the steps
    ω1 := (ψ1 + a21Hy2)/τ1
    ( ψ1
      y2 ) := ( ψ1 - ω1
                y2 - ω1u21 )
    ( ATL | ATR
      ABL | ABR ) ← ( A00 a01 A02
                      a10T α11 a12T
                      A20 a21 A22 ),
    ( tT
      tB ) ← ( t0
               τ1 ), ( yT
                      yB ) ← ( y0
                              ψ1
                              y2 )
endwhile
```

that computes $Q^H y$ if Q (stored as Householder vectors in A) is $m \times n$.

- ☒ $4mn - 2n^2$
- ☐ $2mn$
- ☐ $2mn^2 - \frac{2}{3}n^3 + 2mn$



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